

Polynomials — MCQ Worksheet

Mathematics • Chapter 2 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. A polynomial of degree 2 is called a:

- (A) Linear polynomial
- (B) Quadratic polynomial
- (C) Cubic polynomial
- (D) Constant polynomial

Q2. The degree of the polynomial $4x^5 - 3x^2 + 7$ is:

- (A) 2
- (B) 3
- (C) 5
- (D) 7

Q3. A linear polynomial has at most ____ zeros:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q4. A quadratic polynomial has at most ____ zeros:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q5. A cubic polynomial has at most ____ zeros:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q6. If α and β are zeros of a quadratic $ax^2 + bx + c$, then $\alpha + \beta =$

- (A) b/a
- (B) $-b/a$
- (C) c/a
- (D) $-c/a$

Q7. For $ax^2 + bx + c$, $\alpha\beta =$

- (A) b/a
- (B) $-b/a$
- (C) c/a
- (D) $-c/a$

Q8. The zeros of $x^2 - 7x + 10$ are:

- (A) 2 and 5
- (B) -2 and -5
- (C) 2 and -5
- (D) -2 and 5

Q9. The sum of zeros of $x^2 - 6x + 8$ is:

- (A) 6
- (B) -6
- (C) 8
- (D) -8

Q10. Product of zeros of $2x^2 - 5x + 3$ is:

- (A) $3/2$
- (B) $-3/2$
- (C) $5/2$
- (D) $-5/2$

Q11. Sum of zeros of $2x^2 - 5x + 3$ is:

- (A) $5/2$
- (B) $-5/2$
- (C) $3/2$
- (D) $-3/2$

Q12. A quadratic polynomial whose zeros are 3 and -2 is:

- (A) $x^2 - x - 6$
- (B) $x^2 + x - 6$
- (C) $x^2 + 5x - 6$
- (D) $x^2 - 5x + 6$

Q13. If $\alpha + \beta = 6$ and $\alpha\beta = 9$, the quadratic polynomial is:

- (A) $x^2 - 6x + 9$
- (B) $x^2 + 6x + 9$
- (C) $x^2 - 9x + 6$
- (D) $x^2 - 9x - 6$

Q14. Number of zeros of $f(x) = x^2 + 1$ (real) is:

- (A) 0
- (B) 1
- (C) 2
- (D) Infinite

Q15. A quadratic graph that does not cross x-axis has zeros:

- (A) Two real and equal
(C) No real zeros

- (B) Two real and distinct
(D) Three

Q16. If one zero of $x^2 - kx + 12$ is 4, then $k =$

- (A) 3
(C) 7

- (B) 4
(D) 12

Q17. If α, β, γ are zeros of $ax^3 + bx^2 + cx + d$, then $\alpha + \beta + \gamma =$

- (A) b/a
(C) c/a

- (B) $-b/a$
(D) $-d/a$

Q18. For cubic $ax^3 + bx^2 + cx + d$, $\alpha\beta + \beta\gamma + \gamma\alpha =$

- (A) b/a
(C) $-c/a$

- (B) c/a
(D) $-d/a$

Q19. For cubic $ax^3 + bx^2 + cx + d$, $\alpha\beta\gamma =$

- (A) b/a
(C) $-c/a$

- (B) c/a
(D) $-d/a$

Q20. The graph of $y = x^2$ is a:

- (A) Straight line
(C) Parabola opening downward

- (B) Parabola opening upward
(D) Circle

Q21. If 2 is a zero of $x^2 + ax - 6$, then $a =$

- (A) 1
(C) 2

- (B) -1
(D) 3

Q22. Zeros of $x^2 - 9$ are:

- (A) 3, -3
(C) 0, 9

- (B) 9, -9
(D) Only 3

Q23. If α and β are zeros of $p(x)$, then $p(\alpha) + p(\beta) =$

- (A) 0
(C) $\alpha + \beta$

- (B) 1
(D) $\alpha\beta$

Q24. A polynomial in one variable of degree 0 is:

- (A) A linear polynomial
(C) Zero polynomial

- (B) A constant
(D) Quadratic

Q25. Number of real zeros of $x^3 + x$ is:

- (A) 3
(C) 1

- (B) 2
(D) 0